OMC at TREC 2026 AutoJudge: Methods and Results

Conference notebook paper

Max Irwin* Nathan Day† Daniel Worley‡

10 November 2026

Abstract

Decision models offer a novel (and cheaper) alternative to the established LLM-as-a-judge paradigm (Zheng et al. 2023). Jev, the first decision model publicly released in September 2026, changes the output of a large pre-trained model from the free-form-text of LLMs to a type-safe-selection from a pre-defined set of options. Jev also emits a confidence score for each of its decisions and it offers all of this at an order of magnitude cheaper inference costs. Our submissions explore Jev's performance as a both pair-wise-response judge and one-shot-QREL judge. We also explore a prompt enrichment strategey known as 'bag-of-decisions' leveraging a traditional LLM to build an "enriched" request for Jev grading.

1. Introduction

LLM-as-a-judge has made relevance assesement cheaper than ever. While LLM-judges have become more common in industry applications and create compelling output judgements, they still exhibit major drawbacks that limit their functional utility including: positional-bias (Wang et al. 2024), self-preference (Panickssery et al. 2024), score-centrality and over-confidence. Methods to address these short comings often involed spending more time "reasoning", running more permutation-prompts per single assesment, running several foundations LLMs in a judge-panel, all of which add to expense and over-confidence as context is expanded. We explore the utility of a 'bag of decisions' context enrichment step that uses a separate LLM to build a evaluation rubric of specific yes/no questions that are then used to subsequent Jev prompt.

Jev is a decision model that takes advantage of similar web-scale pre-training like an LLM, but instead of generating free-form-text output, it instead produces an decision based on a prompt-defined rubric. Unlike LLM-defined-rubrics Jev's training regime makes hallucination impossible and rubric adherence absolute. Jev supports making three classes of decision: Noul, Choice and Score. All of which are accompanied by a confidence score.

We also propose two smaller model strategies to be used alongside larger models. The first is Hallucination-Risk (HR), which works independent of LLM or Jev using spaCy (Honnibal and Montani 2017) noun-phrase detection to build an index of concepts for each topic that allows foreground/background calculations for a given run. The second we include is miniLM's (Wang et al. 2020) cosine similarity between a topic and response, which gives us a useful marker for "easy" compared to "hard" examples.

2. Approach

Our approaches are aimed at measuring Jev's utility as a judge, at both levels of RAG relevance assesment. We judge at both the response and the citation level to compare the quality of the assesment to the cost. Along the

*max@onemorecloud.com
†nday@eezy.com
‡dworley@gmail.com

way we explore and try to mitigate Jev's suseptibility to the same LLM issues of: positional-bias, over-confidence and score-centrality. Our run naming convention is confusing and we are sorry about that. All of our work is run via OpenRouter and costs are reported in USD ($).

2.1 Pairwise Jev

Pairwise Jev is a run response level judge. Pairwise Jev decides on a preferred response from a provided pair of different run responses. A single prompt asks a NOUL question that includes the topic and two run responses [appendix1]. We run a single prompt twice to capture both orderings of the run responses [A,B] and [B,A] and then average the resulting scores to get a single preference score normalized for position-bias. Bradley-Terry (Bradley and Terry 1952) combines all pair wise comparisons for a given run into a global win/loss score across all topics, where more relevant runs score higher.

2.2 Bag of Decisions rubric enrichment

Bag of decisions is a separate LLM prompt that attempts to enrich the provided topic information beyond just the prompt-statement and information-need provided. Instead of the singular is this response better style question, BOD resquests a list of 5 - 15 rubric questions an expert would ask to check distinct parts of the repsonse for validity. This list is filtered for near-duplicates and then packaged as individual NOUL questions in a new enriched-prompt for the Pairwise Jev, with the final prompt answer scores averaged together before combining both order-permutation versions.

2.2.1 Hallucination Risk Hallucination risk measures a lexical deviation from concepts in the source cited documents to the answer summary text. Using a spaCy large model, we gathered all normalized noun phrases from the cited documents, and gathered all normalized noun phrases from the answers. If an answer used a noun phrase not present in the cited source, it counts as a potential hallucination. Hallucination risk was not submitted as its own judge, but instead was used as a discount for BOD. This discount didn't result in significant change to the leaderboard, but did flip the top two results.

2.3 Judge Jev

Judge-Jev is a QREL level judge, It takes single cited document and the associated topic as input and outputs a relevance score ranging from 0 (not relevant) - 3 (most relevant), following TREC multi-label convention. A single JudgeJev prompt asks 5 questions about the document's relevance, 1 Noul (binary relevance probability scaled by a factor of 3) 2-Choice (fwd/rev single score selection e.g. [0,1,2,3] and [3,2,1,3]), 2-Score (fwd/rev continuous float score estimation e.g. [0-3] and [3-0]). All five of these scores are averaged together to a final QREL estimate for each citation.

These QRELS were summarized in the style of trec_eval to construct various leaderboard metrics. QRELs were not submitted directly to the track, due to evalution limitations. They are available upon requests of the authors and open to publishing with the proceedings.

3. Submissions

[summary table of runs and costs]

Item	run_ids	Cost (USD)
jev pairwise	omc26_ragtime26_jevpairwise, omc26_rag26_jevpairwise	$93.84
jev + BOD	omc26_ragtime26_jevbod, omc26_rag26_jevbod	$1.57

Item	run_ids	Cost (USD)
jev + BOD + HR	omc26_ragtime26_jevbodrisk, omc26_rag26_jevbodrisk	$0.00
JudgeJev	omc26_ragtime26_judge_jev, omc26_rag26_judge_jev	$14.40

3.1 Pairwise Jev Order bias

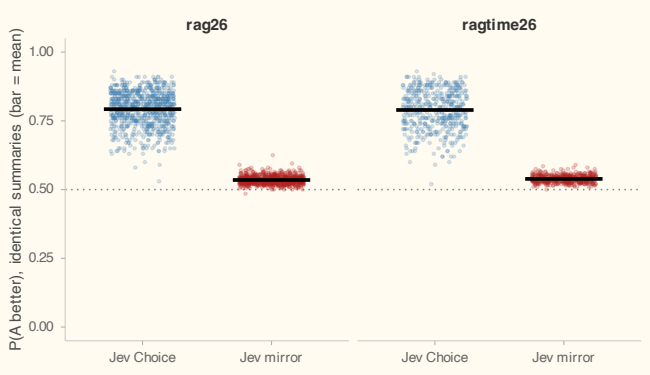


Figure 1: Jev Choice questions exhibit clear order bias in the Pairwise Jev Judge. Using a mirrored version of the choies with order inverted removes most of that bias.

3.2 Pairwise Jev vs BOD Jev + BOD+HR Jev

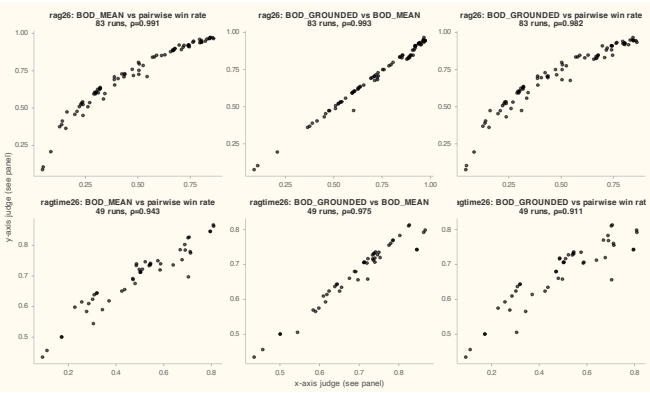


Figure 2: Comparison of leaderboards created with different Jev Judges. Hallucination Risk (Grounded) makes very little change compared to Bag of Decisions (BOD)

3.2 Judge Jev datum

There is strong answer correlation across differen question framing within the same prompt. There are distinc confidence distributions associtate with SCORE and CHOICE answers.

4. Conclusions

After conference.

5. Appendix

5.1 OpenRouter Prompts

5.1.1 Pairwise Jev

```
prompt = '
{
  "model": "typesafe/jev-1.13",
  "state": {
    "query": "fall leaf color change chlorophyll",
    "problem_statement": "My 5-year-old keeps noticing that leaves turn red, orange, and yellow in the fa
    "background": "This is for a child who loves collecting leaves on walks and sorting them by color and
    "summary_a": "Leaves change color in fall because of a green pigment called chlorophyll. When days ge
    "summary_b": "Trees have green stuff in their leaves called chlorophyll. In fall, the green goes away
  },
  "questions": {
    "better_summary": {
      "type": "choice",
      "instructions": "You are a subject matter expert. Two RAG summaries (summary_a and summary_b) were
      "criteria": {
        "A": "summary_a better satisfies the information need of the query.",
        "B": "summary_b better satisfies the information need of the query."
      }
    }
  }
}
'

cat(toJSON(fromJSON(prompt), auto_unbox = TRUE, pretty = TRUE))
```

5.1.2 Bag of Decisions Jev

LLM Bag of Decisions generation

Jev Bag of Questions assembly

5.1.3 Judge Jev

References

Bradley, Ralph Allan, and Milton E. Terry. 1952. "Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons." *Biometrika* 39 (3/4): 324–45. https://doi.org/10.2307/2334029.

Honnibal, Matthew, and Ines Montani. 2017. *spaCy 2: Natural Language Understanding with Bloom Embeddings, Convolutional Neural Networks and Incremental Parsing.* https://spacy.io.

Panickssery, Arjun, Samuel R. Bowman, and Shi Feng. 2024. "LLM Evaluators Recognize and Favor Their Own Generations." *Advances in Neural Information Processing Systems (NeurIPS 2024).* https://arxiv.org/abs/2404.13076.

Wang, Peiyi, Lei Li, Liang Chen, et al. 2024. "Large Language Models Are Not Fair Evaluators." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024)*, 9440–50. https://arxiv.org/abs/2305.17926.

Wang, Wenhui, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers." *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. https://arxiv.org/abs/2002.10957.

Zheng, Lianmin, Wei-Lin Chiang, Ying Sheng, et al. 2023. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. https://arxiv.org/abs/2306.05685.